Fatima Jinnah Women University

DEPARTMENT OF SOFTWARE ENGINEERING

BS(SE) – Semester V

ASSIGNMENT 01 – SOLUTION REPORT

Git, Gitea, GitHub, Git LFS, and GitHub Pages

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Course: Cloud Computing (BSE-410)

Section: A

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Submission Date: 10 October 2026

Environment Setup

Before starting the tasks, the required tools were installed and configured on the Ubuntu server (hostname ubuntu, user irajtaimurmalik). The server runs as a VMware virtual machine with IP address 192.168.28.131.

1. Install Git LFS

```
irajtaimurmalik@ubuntu:~$ sudo apt install -y git-lfs
Installing:
  git-lfs

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 59
  Download size: 4,309 kB
  Space needed: 13.1 MB / 4,786 MB available

Get:1 http://pk.archive.ubuntu.com/ubuntu resolute/universe amd64 git-lfs amd64 3.7.1-1 [4,309 kB]
Fetched 4,309 kB in 4s (1,006 kB/s)
Selecting previously unselected package git-lfs.
(Reading database… 98682 files and directories currently installed.)
Preparing to unpack …/git-lfs_3.7.1-1_amd64.deb…
Unpacking git-lfs (3.7.1-1)…
Setting up git-lfs (3.7.1-1)…
Processing triggers for man-db (2.13.1-1build1)…
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
irajtaimurmalik@ubuntu:~$ git lfs version
git-lfs/3.7.1 (GitHub; linux amd64; go 1.25.0)
```

Figure 1: *Installing Git LFS with apt and checking its version (git-lfs/3.7.1).*

2. Configure Git Identity

```
irajtaimurmalik@ubuntu:~$ git config --global user.name 'Iraj Taimur Malik'
irajtaimurmalik@ubuntu:~$ git config --global user.name
Iraj Taimur Malik
irajtaimurmalik@ubuntu:~$ git config --global user.email 'irajtaimur@gmail.com'
irajtaimurmalik@ubuntu:~$ git config --global user.email
irajtaimur@gmail.com
irajtaimurmalik@ubuntu:~$ _
```

Figure 2: *Setting and verifying the Git user name and email.*

3. Install Docker Engine

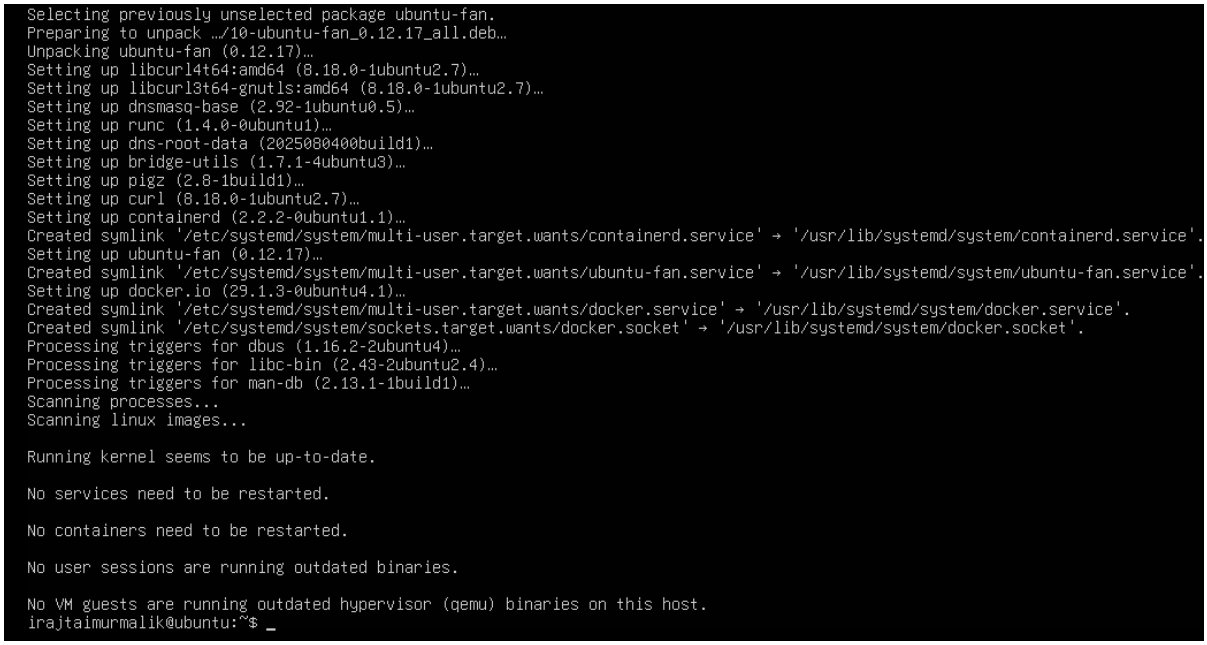


Figure 3: *Installing docker.io and curl on the Ubuntu server.*

4. Docker Compose Plug-in

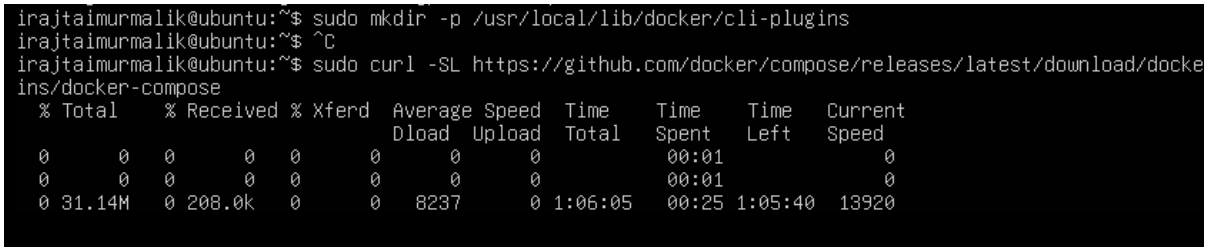


Figure 4: *Creating the Docker CLI plug-in directory and starting the Docker Compose download (the slow download was stopped and Docker Compose was installed from the Ubuntu package repository instead).*

5. Verify Installations

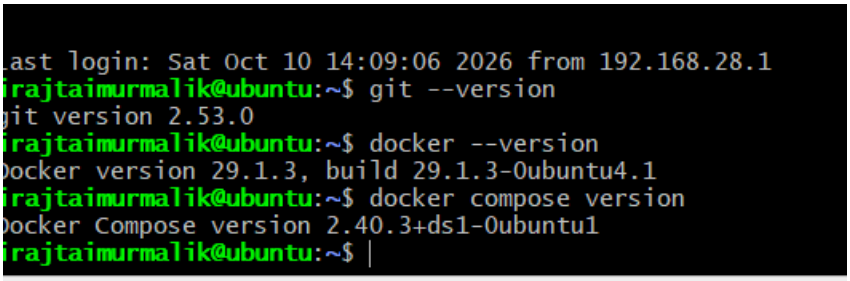


Figure 5: *Versions of Git, Docker, and Docker Compose displayed on the Ubuntu server.*

Task 1: Install Gitea and Push a Repository from the Ubuntu Server

Step 1.1: Install and Run Gitea

The instructor-provided Gitea repository was cloned and Gitea with PostgreSQL was started using the supplied compose.yaml.

```
irajtaimurmalik@ubuntu:~$ cd ~
irajtaimurmalik@ubuntu:~$ git clone https://github.com/WaqasSaleem97/Gitea.git
Cloning into 'Gitea'...
remote: Enumerating objects: 16, done.
remote: Counting objects: 100% (16/16), done.
remote: Compressing objects: 100% (9/9), done.
remote: Total 16 (delta 4), reused 7 (delta 2), pack-reused 0 (from 0)
Receiving objects: 100% (16/16), 5.37 KiB | 1.79 MiB/s, done.
Resolving deltas: 100% (4/4), done.
irajtaimurmalik@ubuntu:~$ cd Gitea
irajtaimurmalik@ubuntu:~/Gitea$ docker compose up -d
[+] Running 14/20
```

Figure 6: *Cloning the instructor's Gitea repository and starting the containers with docker compose up -d.*

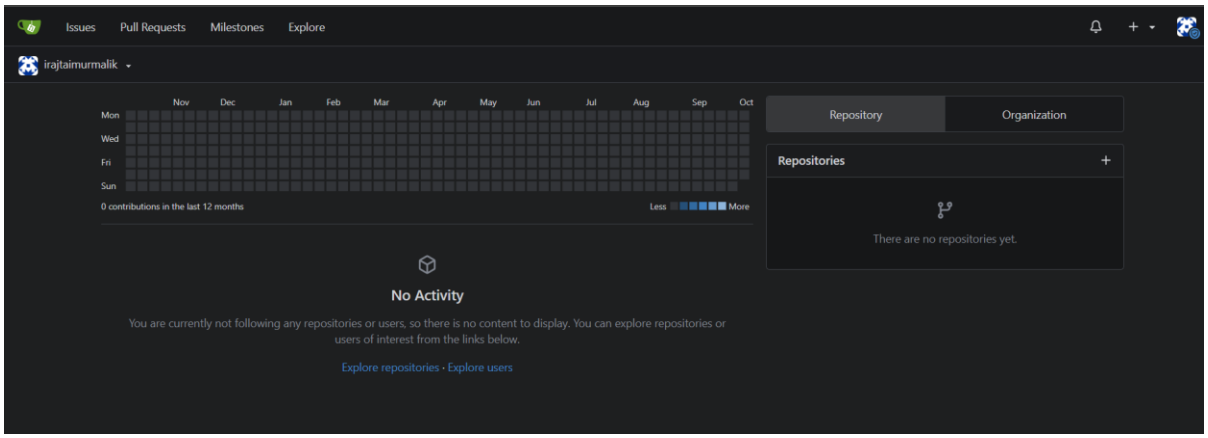


Figure 7: *Gitea installation completed and the administrator account (irajtaimurmalik) logged in.*

```
irajtaimurmalik@ubuntu:~/Gitea$ docker compose ps
NAME       IMAGE                COMMAND                  SERVICE   CREATED          STATUS          PORTS
gitea      gitea/gitea:latest   "/usr/bin/entrypoint…"   gitea     18 minutes ago   Up 18 minutes   22/tcp, 0.0.0.0:3000->3000/tcp, [::]:3000->3000/tcp
gitea_db   postgres:17-alpine   "docker-entrypoint.s…"   db        18 minutes ago   Up 18 minutes   5432/tcp
irajtaimurmalik@ubuntu:~/Gitea$ curl -sS -o /dev/null -w "Gitea HTTP status: %{http_code}\n" http://127.0.0.1:3000/
Gitea HTTP status: 200
irajtaimurmalik@ubuntu:~/Gitea$
```

Figure 8: *docker compose ps showing the Gitea and PostgreSQL containers running, with Gitea HTTP status 200 (task1_gitea_running.png).*

Step 1.2: Create and Push the Initial Repository

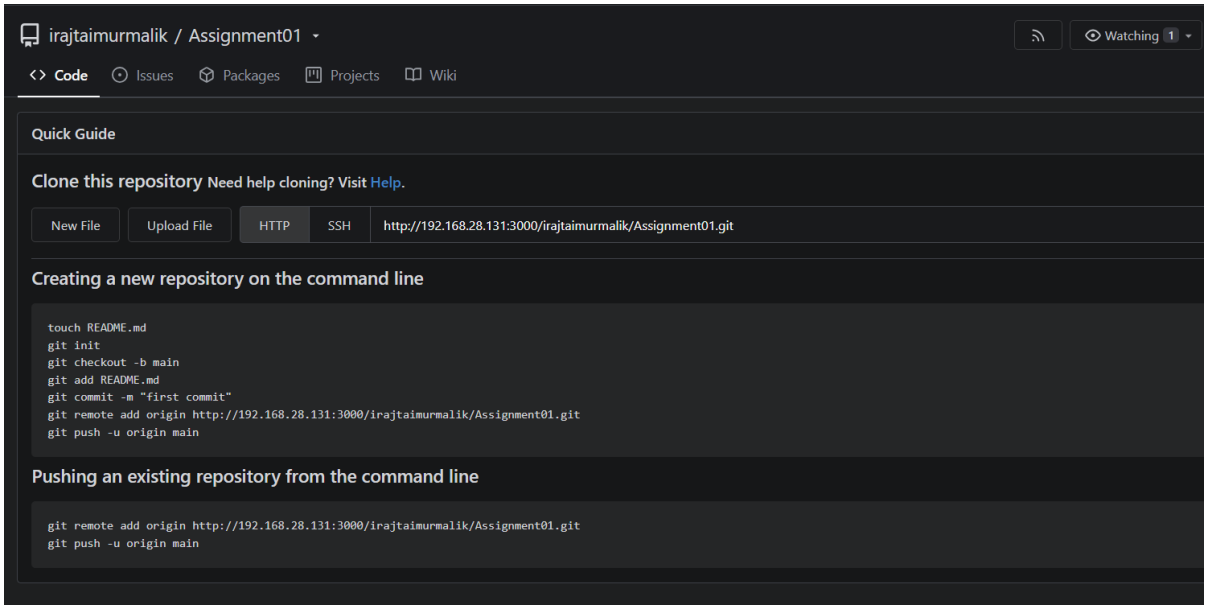


Figure 9: *Empty public Assignment01 repository created in Gitea.*

```
irajtaimurmalik@ubuntu:~/Gitea$ cd ~
irajtaimurmalik@ubuntu:~$ mkdir Assignment01
irajtaimurmalik@ubuntu:~$ cd Assignment01
irajtaimurmalik@ubuntu:~/Assignment01$ git init
hint: Using 'master' as the name for the initial branch. This default branch name
hint: will change to "main" in Git 3.0. To configure the initial branch name
hint: to use in all of your new repositories, which will suppress this warning,
hint: call:
hint:
hint:   git config --global init.defaultBranch <name>
hint:
hint: Names commonly chosen instead of 'master' are 'main', 'trunk' and
hint: 'development'. The just-created branch can be renamed via this command:
hint:
hint:   git branch -m <name>
hint:
hint: Disable this message with "git config set advice.defaultBranchName false"
Initialized empty Git repository in /home/irajtaimurmalik/Assignment01/.git/
irajtaimurmalik@ubuntu:~/Assignment01$ git branch -M main
irajtaimurmalik@ubuntu:~/Assignment01$ nano README.md
irajtaimurmalik@ubuntu:~/Assignment01$ git add README.md
irajtaimurmalik@ubuntu:~/Assignment01$ git commit -m "Add student information"
[main (root-commit) 20c50cf] Add student information
 1 file changed, 3 insertions(+)
 create mode 100644 README.md
```

Figure 10: *Creating the local Assignment01 repository, adding README.md, and committing it.*

```
irajtaimurmalik@ubuntu:~/Assignment01$ git push -u gitea main
Password for 'http://irajtaimurmalik@127.0.0.1:3000':
Enumerating objects: 3, done.
Counting objects: 100% (3/3), done.
Delta compression using up to 2 threads
Compressing objects: 100% (2/2), done.
Writing objects: 100% (3/3), 301 bytes | 150.00 KiB/s, done.
Total 3 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
To http://127.0.0.1:3000/irajtaimurmalik/Assignment01.git
 * [new branch]      main -> main
branch 'main' set up to track 'gitea/main'.
irajtaimurmalik@ubuntu:~/Assignment01$
```

Figure 11: *Pushing the main branch to the gitea remote using the personal access token as the password (task1_gitea_push.png).*

Step 1.3: Verify the Gitea Repository

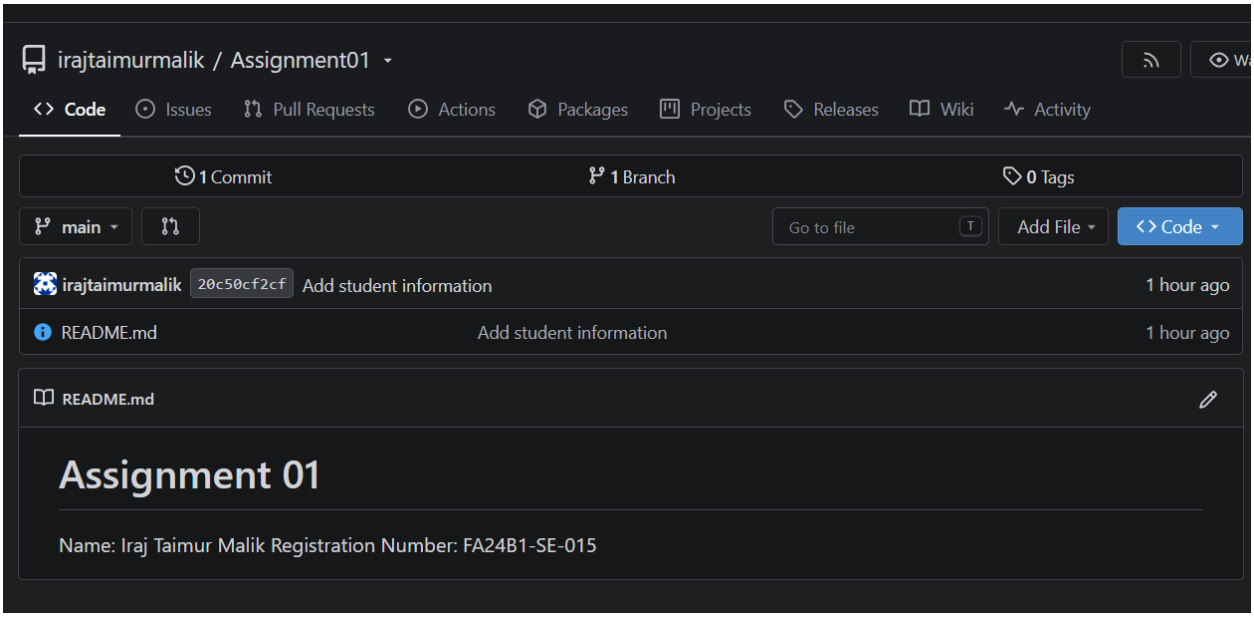


Figure 12: *Gitea Assignment01 repository showing the owner, public status, and rendered README (task1_gitea_repository.png).*

Task 2: Push the Same Repository to GitHub

Step 2.1: Add the GitHub Remote and Verify Both Remotes

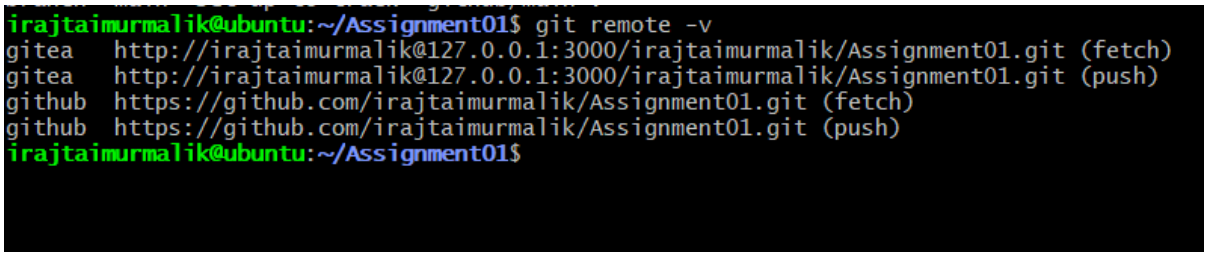


Figure 13: *git remote -v showing both the gitea and github remotes with no token in the URLs (task2_remotes.png).*

Step 2.2: Push to GitHub

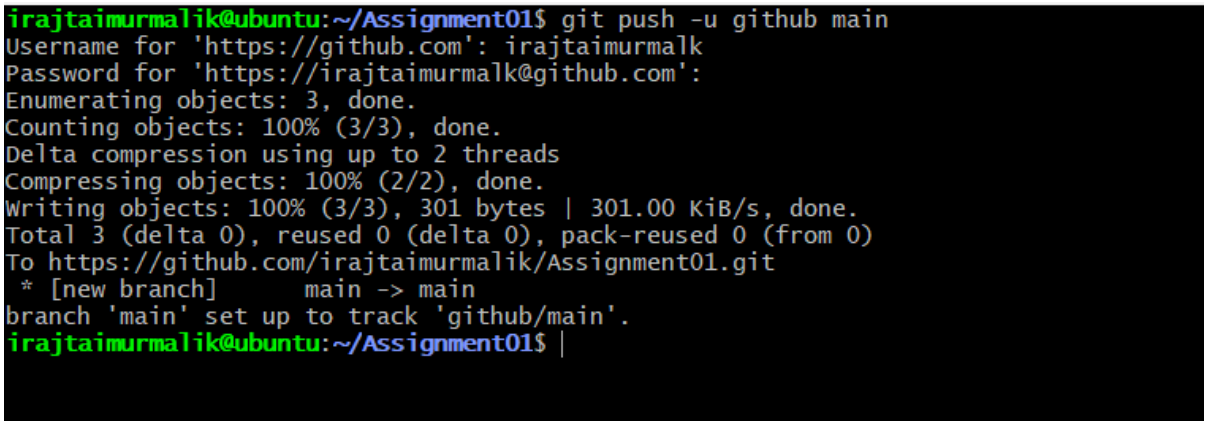


Figure 14: *Pushing the main branch to the github remote (task2_github_push.png).*

Step 2.3: Verify the GitHub Repository

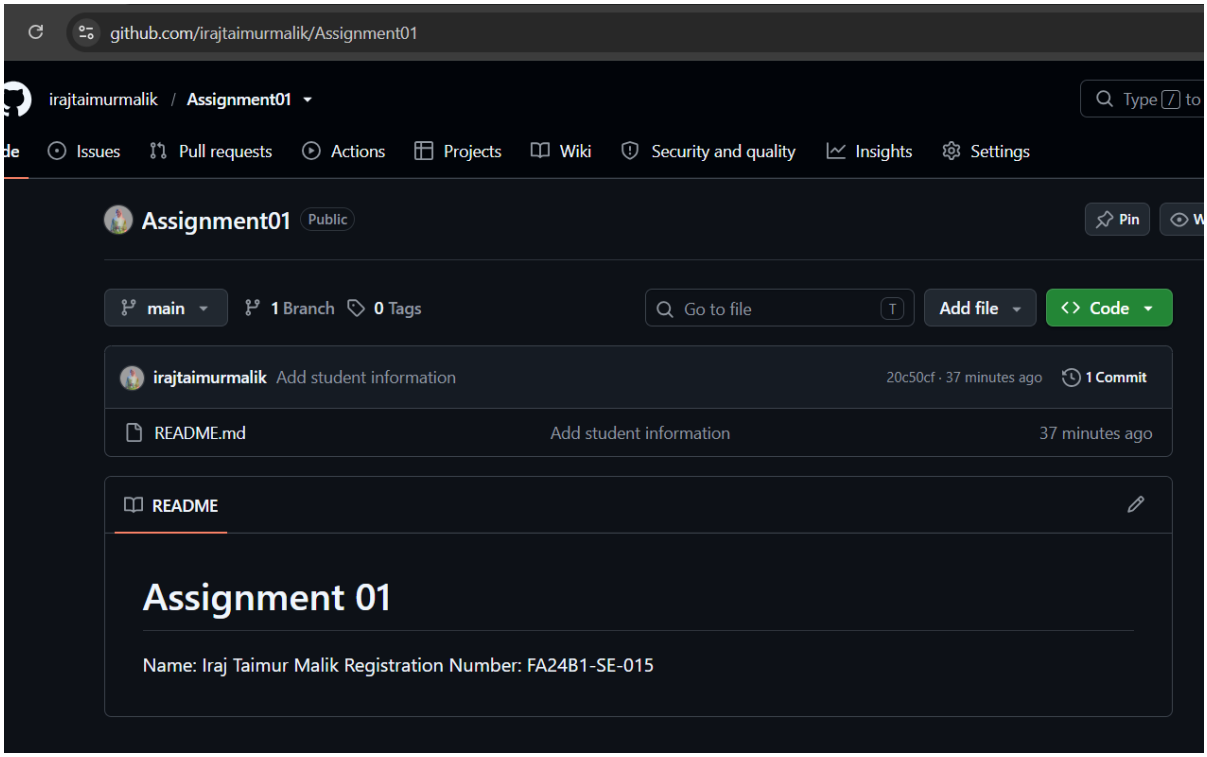


Figure 15: *Public GitHub repository irajtaimurmalik/Assignment01 with the rendered README (task2_github_repository.png).*

Task 3: Track Three Large Files with Git LFS

Step 3.1: Configure Git LFS

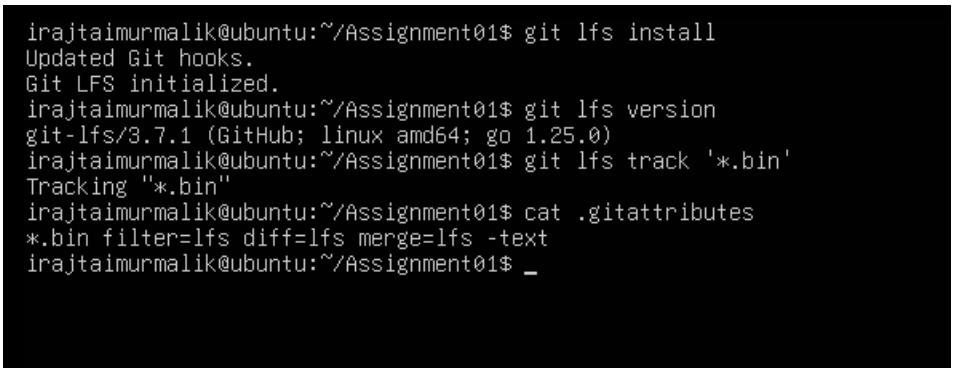


Figure 16: *Initializing Git LFS, checking the version, tracking *.bin files, and the resulting .gitattributes rule (task3_lfs_setup.png).*

Step 3.2: Create and Verify Three LFS Files

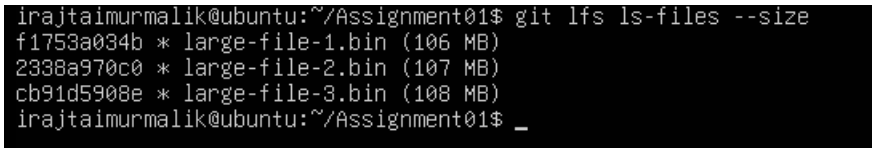


Figure 17: *Three files larger than 100 MiB listed by git lfs ls-files --size (task3_lfs_files.png).*

Task 4: Create a Portfolio or CV with GitHub Pages

Step 4.1: Create and Push the Portfolio Repository

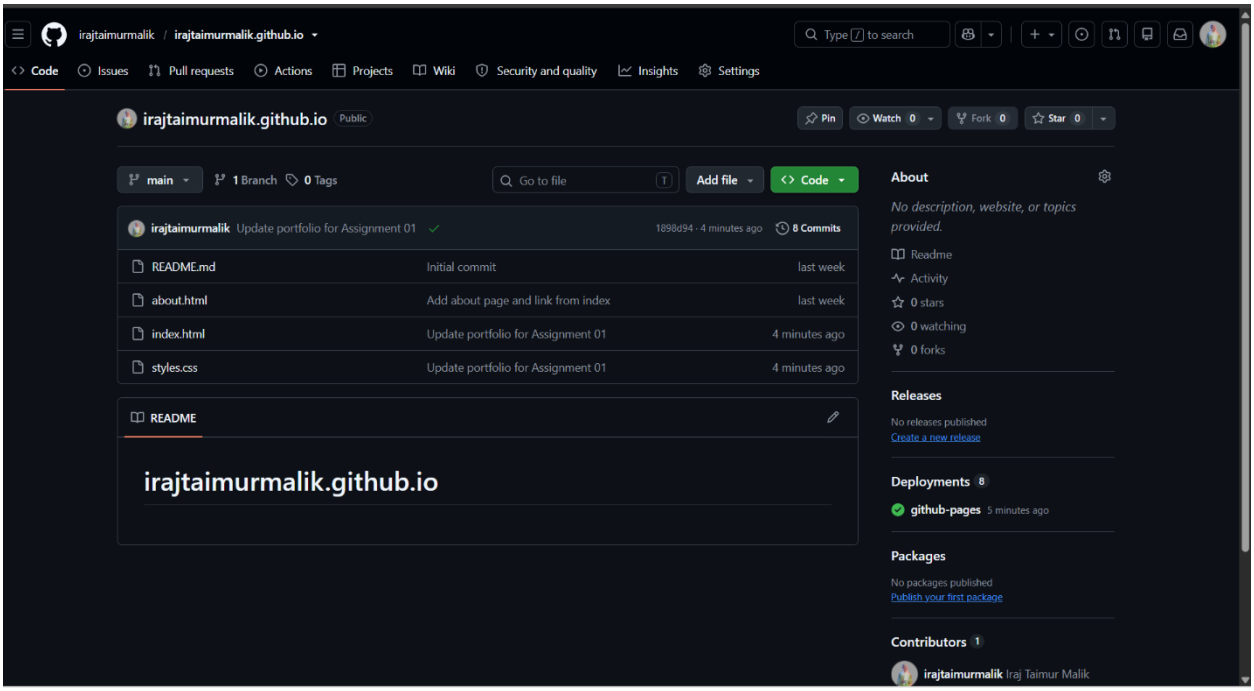


Figure 19: *Repository irajtaimurmalik.github.io containing index.html and styles.css (task4_pages_repository.png).*

Step 4.2: Publish with GitHub Pages

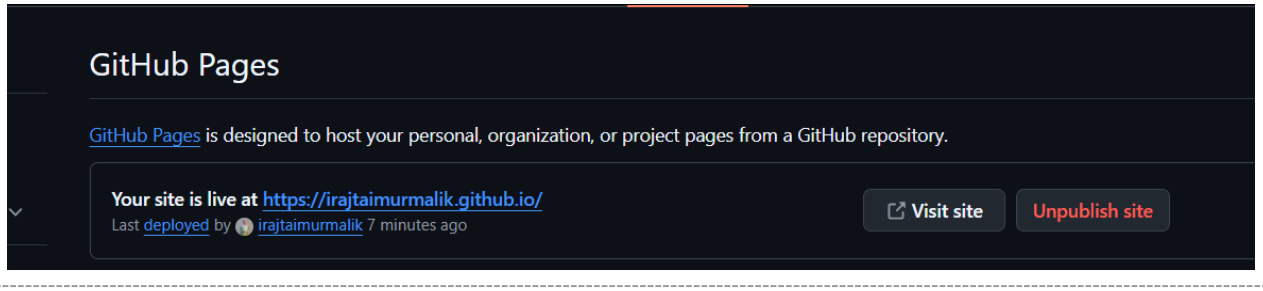
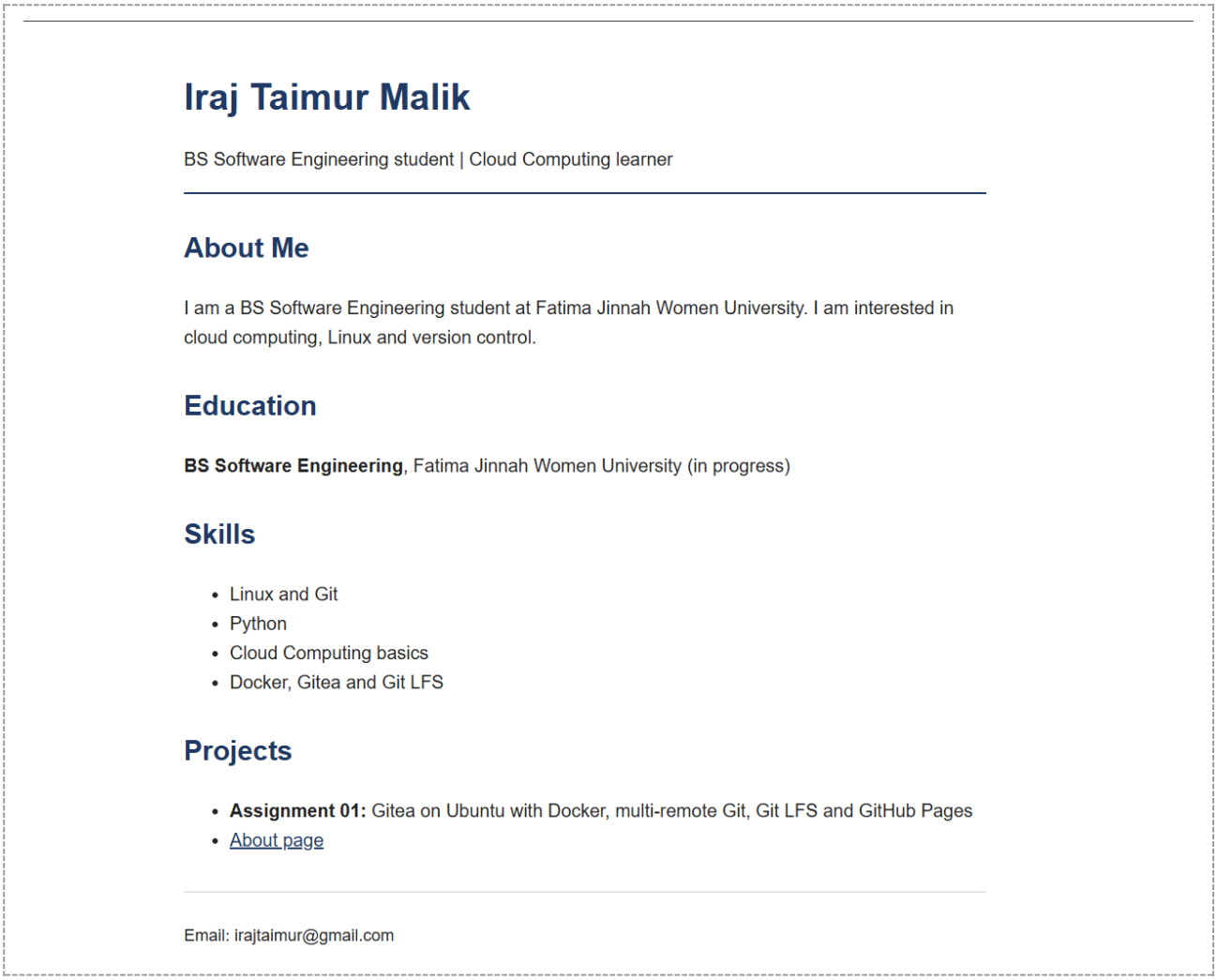


Figure 20: *GitHub Pages settings showing the site is live (task4_pages_deployment.png).*

Step 4.3: Verify the Live Portfolio



Iraj Taimur Malik

BS Software Engineering student | Cloud Computing learner

About Me

I am a BS Software Engineering student at Fatima Jinnah Women University. I am interested in cloud computing, Linux and version control.

Education

BS Software Engineering, Fatima Jinnah Women University (in progress)

Skills

- Linux and Git
- Python
- Cloud Computing basics
- Docker, Gitea and Git LFS

Projects

- **Assignment 01:** Gitea on Ubuntu with Docker, multi-remote Git, Git LFS and GitHub Pages
- About page

Email: irajtaimur@gmail.com

Figure 21: *Live portfolio at https://irajtaimurmalik.github.io/ (task4_portfolio_live.png).*

Submission Links

GitHub Assignment01 repository: https://github.com/irajtaimurmalik/Assignment01

Portfolio (GitHub Pages): https://irajtaimurmalik.github.io/

CC repository: https://github.com/irajtaimurmalik/CC